

RIFAT HOSSAIN TALUKDER

TRANSPORTATION & INFRASTRUCTURE | GIS & REMOTE SENSING | CLIMATE RESILIENCE & SPATIAL MODELLING

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RESEARCH PROFILE

Urban and rural planning researcher working at the intersection of transportation, infrastructure-led change, climate resilience and geospatial analysis. Experience includes mixed-methods counterfactual research, GIS and Earth observation, Random Forest modelling, spatial vulnerability assessment, qualitative research and scenario-based planning. Current research identity is centered on how major infrastructure and environmental change reshape mobility, development patterns, exposure and access.

EDUCATION

Bachelor of Urban and Rural Planning

2018 - 2023

Khulna University, Bangladesh

CGPA: 3.42/4.00 | **Undergraduate thesis:** "Predicting the Influence of the Padma Bridge on Future Urban Growth."

PUBLICATIONS AND MANUSCRIPT

Conference proceedings paper. Himel, T. I., Talukder, R. H., Hossain, M. Z., Naz, S., Arfin, S., & Fariha, J. N. "Assessing Spatial Flood Vulnerability of Greater Rangpur Region in Bangladesh Using GIS: A Tool for Resilience Planning at Regional Scale." In *Planning for Inclusiveness and Sustainability in Post Pandemic Era* (pp. 228-244). Bangladesh Institute of Planners. ISBN 978-984-35-2613-7.

Manuscript under review. Talukder, R. H., Rahman, M. A., Arfin, S., Fariha, J. N., Husna, A. U., Ha-mim, N. M., Hossain, M. Z., & Rahaman, K. R. "A Principal Components Analysis (PCA) Based Novel Approach for Assessing Climate Change Induced Loss and Damage in the Coastal Region of Bangladesh." *The Journal of Environment & Development* (manuscript under review).

RESEARCH AND TECHNICAL EXPERIENCE

Research Assistant (part-time)

Jun 2021 - Dec 2024

University Grants Commission of Bangladesh

- Applied GIS, land-cover analysis and historical-spatial research to campus development, urban land-use change, environmental conditions and the physical evolution of Dhaka.
- Collected, digitized and interpreted historical maps, planning documents, photographs and archival records, with particular attention to the Ramna area and University of Dhaka campus.
- Integrated spatial evidence with building histories, construction information and conservation records to produce maps, timelines, technical documentation and analytical narratives.

Intern, Sundarbans Area Based Programme

Feb 2023 - Jul 2023

United Nations Development Programme (UNDP) | Khulna, Bangladesh

- Designed a remote-sensing and machine-learning workflow to predict cyclone-induced flood vulnerability in Khulna, Satkhira and Bagerhat using historical cyclone scenarios and satellite-derived environmental indicators.
- Prepared Landsat-based indices, trained Random Forest regression models, evaluated predictive performance and translated outputs into flood-vulnerability maps for disaster-risk and resilience planning.
- Conducted literature review, geospatial data preparation, model interpretation and technical reporting on Cyclones Sidr and Aila and their environmental impacts.

Qualitative Researcher

Aug 2023 - Oct 2023

Tiller Bangladesh | Dhaka, Bangladesh

- Designed and conducted interviews, focus-group discussions and key-informant interviews on fishing practices, socioeconomic conditions and conservation knowledge.
- Transcribed, coded and interpreted qualitative data, identified recurring themes and contributed to evidence-based research reports while maintaining confidentiality and research ethics.

Technical Consultant

Mar 2021 - Jun 2021

Centre for Environmental and Geographic Information Services (CEGIS) | Dhaka, Bangladesh

- Digitized census-related spatial datasets and developed GIS layers for administrative boundaries, settlements, infrastructure and socioeconomic indicators.
- Performed spatial-data quality control for scale, positional accuracy, consistency and formatting to support reliable integration across datasets.

SELECTED RESEARCH PROJECTS AND EVIDENCE

Deciphering Traffic Dynamics: Analysing the Induced Traffic of the Padma Multipurpose Bridge Transport research ongoing

Ongoing research

- Used a mixed-methods design combining temporal and spatial no-bridge counterfactual analysis, qualitative behavioural evidence, Monte Carlo uncertainty analysis and a 30-year scenario forecast.
- For the analysed count period, observed bridge traffic was 20,240 vehicles/day versus a preferred temporal no-bridge estimate of 8,052 vehicles/day, yielding 12,188 vehicles/day of bridge-attributable excess traffic (60.2% of observed traffic).
- Monte Carlo analysis produced a median excess estimate of 11,696 vehicles/day; qualitative evidence indicated increased trip frequency and newly feasible travel alongside route diversion and mode transfer.
- Kept bridge-attributable excess traffic conceptually separate from pure induced demand and treated the 2054 forecast range (33,169-44,537 vehicles/day) as scenario-based planning evidence rather than a single deterministic prediction.

Predicting Cyclone-Induced Flood Vulnerable Areas in Coastal Bangladesh

Original study 2023;
extension 2026 -
present

Completed UNDP study with ongoing methodological extension

- Built the original study around Khulna, Satkhira and Bagerhat, using Cyclone Sidr as a historical training context and Cyclone Aila as a cross-event test.
- Combined Landsat 5/8/9 imagery with GNDVI, NDVI, NDWI, land-surface temperature, salinity index and simple-ratio indicators; used temporal NDWI change as a proxy for flood intensity and Random Forest regression to model nonlinear environmental relationships.
- Evaluated performance using R-squared, MAE and RMSE; the original analysis reported R² values of 0.83 for the Sidr scenario and 0.79 for the Aila scenario.
- Current extension revisits temporal validation, geospatial inputs and a broader vulnerability framework incorporating land use, infrastructure and localized resilience factors.

Assessing Spatial Flood Vulnerability of Greater Rangpur Region in Bangladesh Using GIS

Conference
proceedings · 2021

Published collaborative research

- Co-authored a regional GIS-based assessment integrating physical and socioeconomic indicators to support resilience-oriented planning in Greater Rangpur.

Predicting the Influence of the Padma Bridge on Future Urban Growth

Undergraduate thesis ·
completed 2023

Completed undergraduate research

- Used multi-date Landsat imagery, Random Forest classification, spatial predictors, Markov transition modelling and Cellular Automata to explore infrastructure-associated urban-growth scenarios.
- Developed the foundation for later research in transport geography, accessibility, induced demand, land-use change and infrastructure-led urban transformation.

University of Dhaka historical-spatial research

Jan 2025 - present

Active research

- Examining the planning, construction and conservation of University of Dhaka buildings (1921-2018) through historical mapping, archival documents, building histories and spatial reconstruction.

PROPOSED RESEARCH DIRECTIONS

Climate-resilient mobility and healthcare accessibility in Dhaka: concept-stage network-based GIS research on flood disruption, travel time and access to essential services.

Multi-hazard and infrastructure vulnerability in coastal Bangladesh: concept-stage multi-source geospatial research combining satellite observations, terrain, land use, climate and infrastructure information.

Green exposure, accessibility and spatial inequality in Dhaka: concept-stage neighborhood analysis of walkability, green exposure, heat and service accessibility.

TEACHING AND ACADEMIC LEADERSHIP

Head & Founding Member, Research and Academic Excellence Department

Jul 2025 - present

IELTS School BD | Dhaka, Bangladesh

- Co-founded and lead a research and academic-support unit serving students preparing for higher education, with responsibility for research mentoring and academic quality assurance.
- Teach research design, literature review, quantitative and qualitative methods, research ethics, academic writing and source evaluation.
- Mentor students in selecting and applying analytical tools, including GIS, remote sensing and statistical software, according to the research question.
- Lead curriculum development, faculty support, assessment review and monitoring of academic outcomes.

Teaching Assistant

Jan 2024 - Jun 2025

Khulna University | Khulna, Bangladesh

- Guided students in cleaning, transforming, integrating and interpreting spatial, transportation and traffic datasets from multiple sources.
- Supported instruction in GIS, remote sensing, transport planning and traffic simulation, emphasizing assumptions, data quality, scenario comparison, limitations and evidence-based interpretation.

SELECTED TECHNICAL AND RESEARCH SKILLS

GIS and Earth observation: ArcGIS Pro, QGIS, ArcPy, Google Earth Engine, ERDAS IMAGINE, Dinamica EGO; spatial analysis; land-use/land-cover analysis; remote-sensing indices; raster/vector processing; cartographic communication.

Geospatial machine learning, programming and data analysis: Python, R, SPSS, Jamovi, NVivo; Random Forest regression; quantitative analysis; qualitative coding; data visualization; model evaluation using R², MAE and RMSE.

Transportation and spatial modelling: counterfactual analysis; induced-demand research; scenario-based forecasting; uncertainty analysis; network and accessibility concepts; TransCAD, MATSim and VISSIM exposure through research and teaching.

Research methods: research design; literature review; quantitative and qualitative methods; interviews; focus groups; key-informant interviews; thematic analysis; spatial equity analysis; research ethics; academic writing and citation management.

Additional: AutoCAD; technical reporting; archival and historical-spatial research; map-based research communication.

SELECTED ACADEMIC, PLANNING AND COMMUNITY SERVICE

Invited Volunteer Planning Reviewer, Bangladesh Paurashava initiatives

Mar 2022 - Dec 2022

Reviewed selected local planning and development proposals by invitation, contributing observations on land use, accessibility, public facilities and spatial planning.

Voluntary Executive, GIS Club, Khulna University

Jan 2021 - Dec 2022

Supported peer learning and student activities related to GIS, remote sensing, mapping and spatial thinking.

Volunteer, Help the Children

Aug 2020 - Dec 2021

Contributed to child-focused educational support, learning assistance and community outreach.

Blood Donor, Badhon - Khulna University

Jun 2018 - present

Participate in voluntary blood-donation and student-led community service activities.

Student Volunteer, 21st N-AERUS Conference

2022

Supported the conference "Transforming Role of State in Addressing Inclusive Urban Development" in Khulna and gained exposure to international academic discussions on inclusive urban development.

SELECTED HONORS AND AWARDS

Top Performer, Machine Learning Applications in Remote Sensing & GIS Training - Geospatial Planning (2022). Recognized for strong performance in integrating machine-learning approaches with geospatial and remote-sensing workflows.

Certificate of Achievement for Outstanding Contribution to the Betterment of the University and Society - Khulna University (2021).

Second runner-up, Urban Design Project Competition - Bangladesh Institute of Planners (2020).

ENGLISH PROFICIENCY

IELTS (October 25, 2024): **Overall 8.0** | Listening 9.0 | Reading 8.5 | Writing 7.0 | Speaking 7.0.